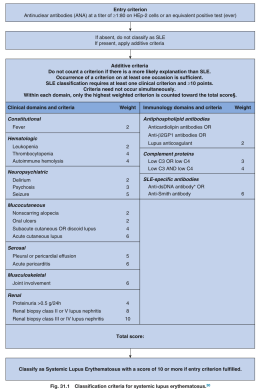


Entry criterion
Antinuclear antibodies (ANA) at a titer of ≥1:80 on HEp-2 cells or an equivalent positive test (once)

If absent, do not classify as SLE
If present, apply additive criteria

Additive criteria
Do not count a criterion if there is a more likely explanation than SLE.
Occurrence of a criterion on at least one occasion is sufficient.
SLE classification requires at least one clinical criterion and ≥10 points.
Criteria need not occur simultaneously.
Within each domain, only the highest weighted criterion is counted toward the total score.

Clinical domains and criteria	Weight
Constitutional	
Fever	2
Hematologic	
Leukopenia	3
Thrombocytopenia	4
Autoimmune hemolysis	4
Neuropsychiatric	
Delirium	2
Psychosis	3
Seizure	5
Mucocutaneous	
Nonscarring alopecia	2
Oral ulcers	2
Subacute cutaneous OR discoid lupus	4
Acute cutaneous lupus	6
Serosal	
Pleural or pericardial effusion	5
Acute pericarditis	6
Musculoskeletal	
Joint involvement	6
Renal	
Proteinuria >0.5 g/24h	4
Renal biopsy class II or V lupus nephritis	8
Renal biopsy class III or IV lupus nephritis	10
Total score:	

Immunology domains and criteria	Weight
Antiphospholipid antibodies	
Anticardiolipin antibodies OR	
Anti-β2GP1 antibodies OR	
Lupus anticoagulant	2
Complement proteins	
Low C3 OR low C4	3
Low C3 AND low C4	4
SLE-specific antibodies	
Anti-dsDNA antibody* OR	
Anti-Smith antibody	6

Classify as Systemic Lupus Erythematosus with a score of 10 or more if entry criterion fulfilled.

Fig. 31.1 Classification criteria for systemic lupus erythematosus.[7]

Fig_31_1

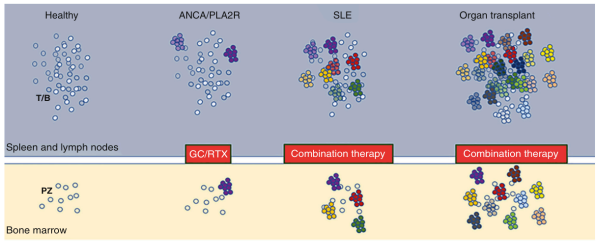


Fig. 31.2 Number of autoreactive T, B, and plasma cell clones in glomerular diseases with single antigen targets, systemic lupus erythematosus, and organ transplant. Immunological activity of all chronic autoimmune diseases is controlled by the size and antigen affinity of autoreactive clones of memory T cells and memory B cells (T/B) in the lymphoid organs as well as by long-lived plasma cells (PZ) in the bone marrow. In "simple" autoimmune diseases with single antigens such as, e.g. proteinase-3 in granulomatous polyangiitis or phospholipase A2 receptor in primary membranous glomerulonephritis, few immunotherapy is needed to control the few clones. In contrast, SLE involves autoimmunity against many, maybe 10-100 antigens, which goes together with a high number of T-, B-, and plasma cell clones to control, which requires combination therapy. Recipients of an organ transplant are in an even worst condition as each transplanted alloantigen will prime such a set of allo-reactive clones, which can go into the 10,000s of clones, difficult to control even with combination therapy. Different auto-reactive T, B, and plasma cell clones are depicted with different colours. *ANCA*, Antineutrophil cytoplasmic antibody; *PLA2R*, phospholipase A2 receptor; *SLE*, systemic lupus erythematosus; *GC*, glucocortcoids; *RTX*, rituximab. (From Parodis I, Depascale R, Doria A, Anders HJ. When should targeted therapies be used in the treatment of lupus nephritis: early in the disease course or in refractory patients? *Autoimmun Rev*. 2024;23(1):103418.)

Fig_31_2

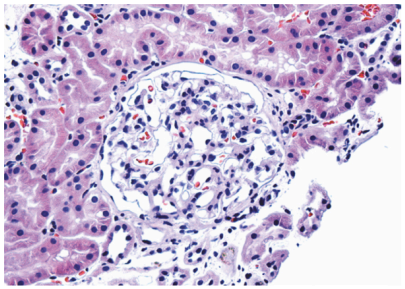


Fig. 31.3 Class II mesangial proliferative lupus nephritis. Increase in mesangial matrix and cellularity. Peripheral capillary loops are normal. The adjacent tubules in the interstitium are uninvolved (hematoxylin-eosin, ×400). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

Fig_31_3

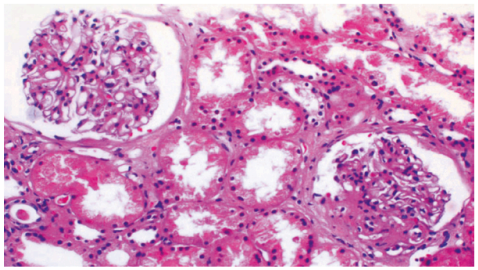


Fig. 31.4 Class III lupus nephritis. Focal and segmental glomerulonephritis. The glomerulus in the upper left pole is normal appearing, whereas the one on the right demonstrates a segmental area of adhesion and necrosis (hematoxylin-eosin, ×400). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

Fig_31_4

Fig 31.5 Class III lupus nephritis. Electron microscopy demonstrating subendothelial and mesangial deposits (transmission electron microscopy, ×4000). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

Fig_31_5

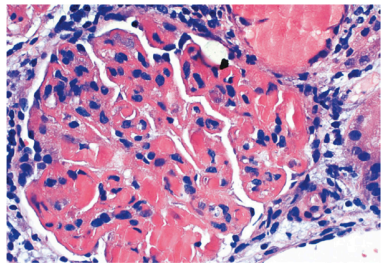


Fig 31.6 Class IV LN. Diffuse proliferative LN involves most or all of the glomeruli. The capillaries here contain "hyaline thrombi," a lesion caused by massive subendothelial deposits bulging into the capillary lumen. Hyaline thrombi may be thickened to the point to be considered "wire-loop" lesions, due to the extensive subendothelial deposits (hematoxylin-eosin, ×400). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

Fig_31_6

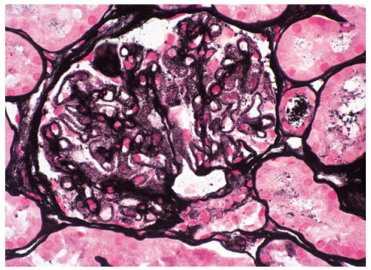


Fig. 31.7 Class V lupus membranous nephritis. Silver methenamine (Jones) stains reveal a spike and dome pattern to be present along the peripheral capillary loops. Where the wall of the capillaries is cut tangentially, there is a moth-eaten appearance of the capillary wall due to the deposits not staining with silver (Jones, ×400). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

Fig_31_7

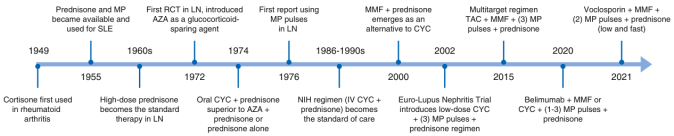


Fig. 31.8 Evolution of immunosuppressive therapies in lupus nephritis.[223]

Fig_31_8

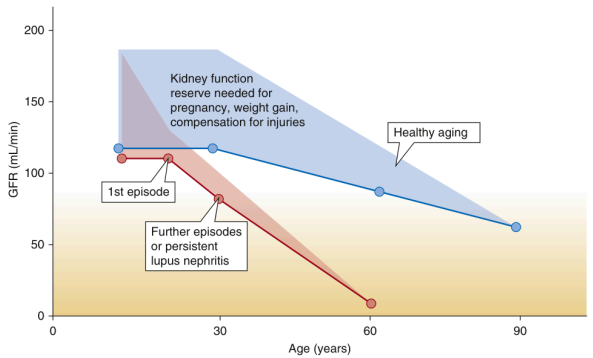


Fig. 31.9 Kidney function decline in healthy adults versus those with lupus nephritis.[39]

Fig_31_9

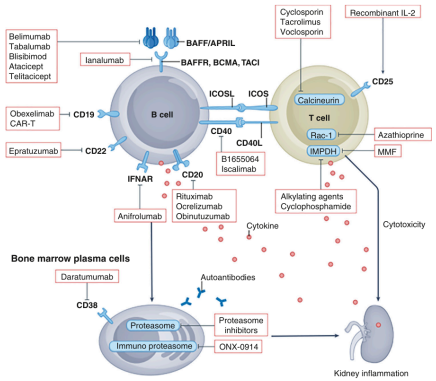
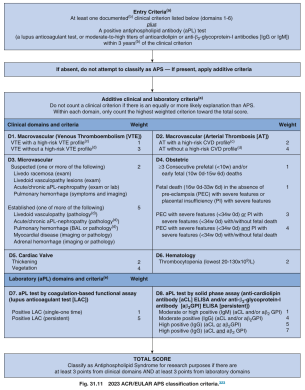


Fig. 31.10 Current targets of investigational therapies in lupus nephritis.[29] B cells are precursors for plasma cells. B cells present autoantigens to T cells and hence drive the expansion of autoantigen-specific T cell clones. For this reason, drug developers primarily invest in B-cell targeting therapies to minimize autoimmune activity in lupus nephritis, which is mediated, among others, by autoantibodies, cytokines and T cell cytotoxicity.

Fig_31_10



Entry Criteria[48]
At least one documented[49] clinical criterion listed below (domains 1-6)
plus
A positive antiphospholipid antibody (aPL) test
(a lupus anticoagulant test, or moderate-to-high titers of anticardiolipin or anti-β_2-glycoprotein-I antibodies [IgG or IgM])
within 3 years[50] of the clinical criterion

If absent, do not attempt to classify as APS — If present, apply additive criteria

Additive clinical and laboratory criteria[48]
Do not count a clinical criterion if there is an equally or more likely explanation than APS.
Within each domain, only count the highest weighted criterion toward the total score.

Clinical domains and criteria	Weight		Weight
D1. Macrovascular (Venous Thromboembolism [VTE])		**D2. Macrovascular (Arterial Thrombosis [AT])**	
VTE with a high-risk VTE profile[51]	1	AT with a high-risk CVD profile[52]	2
VTE without a high-risk VTE profile[51]	3	AT without a high-risk CVD profile[52]	4
D3. Microvascular		**D4. Obstetric**	
Suspected (one or more of the following):	2	≥3 Consecutive prefetal (<10w) and/or	1
Livedo racemosa (exam)		early fetal (10w 0d-15w 6d) deaths	
Livedoid vasculopathy lesions (exam)			
Acute/chronic aPL-nephropathy (exam or lab)		Fetal death (10w 0d-33w 6d) in the absence of	1
Pulmonary hemorrhage (symptoms and imaging)		pre-eclampsia (PEC) with severe features or	
		placental insufficiency (PI) with severe features	
Established (one of more of the following):	5		
Livedoid vasculopathy (pathology[53])		PEC with severe features (<34w 0d) or PI with	3
Acute/chronic aPL-nephropathy (pathology[54])		severe features (<34w 0d) with/without fetal death	
Pulmonary hemorrhage (BAL or pathology[55])		PEC with severe features (<34w 0d) and PI with	4
Myocardial disease (imaging or pathology)		severe features (<34w 0d) with/without fetal death	
Adrenal hemorrhage (imaging or pathology)			
D5. Cardiac Valve		**D6. Hematology**	
Thickening	2	Thrombocytopenia (lowest 20-130x10^9/L)	2
Vegetation	4		
Laboratory (aPL) domains and criteria[50]	**Weight**		
D7. aPL test by coagulation-based functional assay (lupus anticoagulant test [LAC])		**D8. aPL test by solid phase assay (anti-cardiolipin antibody [aCL] ELISA and/or anti-β_2-glycoprotein-I antibody [aβ_2GPI] ELISA [persistent])**	
Positive LAC (single-one time)	1	Moderate or high positive (IgM) (aCL and/or aβ_2 GPI)	1
Positive LAC (persistent)	5	Moderate positive (IgG) (aCL and/or aβ_2GPI)	4
		High positive (IgG) (aCL or aβ_2GPI)	5
		High positive (IgG) (aCL and aβ_2 GPI)	7

TOTAL SCORE
Classify as Antiphospholipid Syndrome for research purposes if there are
at least 3 points from clinical domains AND at least 3 points from laboratory domains

Fig. 31.11 2023 ACR/EULAR APS classification criteria.[54]

Fig_31_11

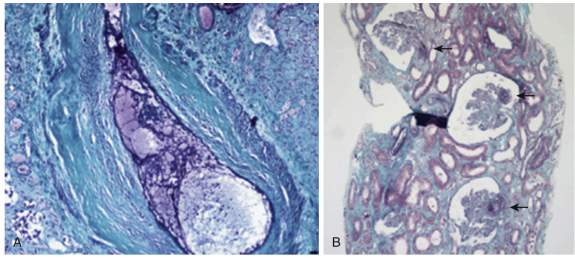


Fig. 31.12 Antiphospholipid nephropathy.[353] (A) Severe fibrous intimal hyperplasia lesion in a native kidney for a patient with APS. (B) Thrombotic microangiopathy lesions in patients with APS *(arrows)*.

Fig_31_12

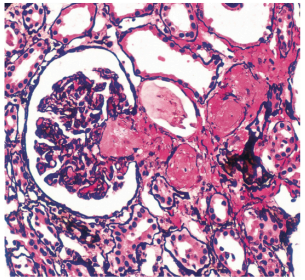


Fig. 31.13 Scleroderma renal crisis. Fibrinoid necrosis of the arteriole is present, with fibrin thrombi occluding the lumen and extending to the hilus of the glomerulus. The glomerulus itself shows only ischemic change with corrugation of the glomerular basement membrane and retraction of the tuft (Jones silver stain, ×200). (From Fogo A, Kashgarian M. *Diagnostic atlas of renal pathology*. 3rd ed. Elsevier, 2017.)

Fig_31_13

Table 31.1 International Society of Nephrology/ Renal Pathology Society (2003) Classification of Lupus Nephritis

Class I Minimal mesangial lupus nephritis (LN)
Class II Mesangial proliferative LN
Class III Focal LN[a] (<50% of glomeruli)
- III (A) Active lesions
- III (A/C) Active and chronic lesions
- III (C) Chronic lesions

Class IV Diffuse LN[a] (≥50% of glomeruli)
Diffuse segmental (IV-S) or global (IV-G) LN
- IV (A) Active lesions
- IV (A/C) Active and chronic lesions
- IV (C) Chronic lesions

Class V[c] Membranous LN
Class VI Advanced sclerosing LN
(≥90% globally sclerosed glomeruli without residual activity)

[a]Indicates the proportion of glomeruli with active and with sclerotic lesions.
[b]Indicates the proportion of glomeruli with fibrinoid necrosis and cellular crescents.
[c]Class V may occur in combination with III or IV, in which case both will be diagnosed.
Indicate and grade (mild, moderate, or severe) tubular atrophy, interstitial inflammation and fibrosis, severity of arteriosclerosis or other vascular lesions.

Table_31_1

Table 31.2 Risks Associated With Cyclophosphamide Use and Mitigation Strategies

Side Effect	Mitigation/Monitoring Strategies
Bone marrow suppression	Regularly monitor complete blood counts (CBC) and adjust dosage as needed in response to leukopenia.
Hemorrhagic cystitis	Ensure adequate hydration and administer mesna to protect the bladder.
Infertility	Discuss fertility preservation options with patients before starting treatment. Ensure cumulative doses do not exceed 10 gm in men and women over the age of 30.
Secondary malignancies	Long-term monitoring for signs of secondary cancers. Ensure secondary screening up to date. Risks of malignancy low, aside from squamous cell cancers, when cumulative doses below 10 gm. Highest risk of solid organ cancers when cumulative doses exceed 36 gm.
Infections	Screen for chronic infections and provide appropriate immunizations (influenza, pneumococcal, SARS-CoV-2) before starting treatment. Consider if prophylactic antibiotics required and regular monitoring for signs of infection.

Table_31_2